

CHAPTER 12—TESTS OF GOODNESS OF FIT AND INDEPENDENCE

MULTIPLE CHOICE

1. A population where each element of the population is assigned to one and only one of several classes or categories is a
 - a. multinomial population
 - b. Poisson population
 - c. normal population
 - d. None of these alternatives is correct.
2. The sampling distribution for a goodness of fit test is the
 - a. Poisson distribution
 - b. t distribution
 - c. normal distribution
 - d. chi-square distribution
3. A goodness of fit test is always conducted as a
 - a. lower-tail test
 - b. upper-tail test
 - c. middle test
 - d. None of these alternatives is correct.
4. An important application of the chi-square distribution is
 - a. making inferences about a single population variance
 - b. testing for goodness of fit
 - c. testing for the independence of two variables
 - d. All of these alternatives are correct.
5. The number of degrees of freedom for the appropriate chi-square distribution in a test of independence is
 - a. $n-1$
 - b. $K-1$
 - c. number of rows minus 1 times number of columns minus 1
 - d. a chi-square distribution is not used
6. In order not to violate the requirements necessary to use the chi-square distribution, each expected frequency in a goodness of fit test must be
 - a. at least 5
 - b. at least 10
 - c. no more than 5
 - d. less than 2
7. A statistical test conducted to determine whether to reject or not reject a hypothesized probability distribution for a population is known as a
 - a. contingency test
 - b. probability test
 - c. goodness of fit test
 - d. None of these alternatives is correct.

8. The degrees of freedom for a contingency table with 12 rows and 12 columns is
 - a. 144
 - b. 121
 - c. 12
 - d. 120
9. The degrees of freedom for a contingency table with 6 rows and 3 columns is
 - a. 18
 - b. 15
 - c. 6
 - d. 10
10. The degrees of freedom for a contingency table with 10 rows and 11 columns is
 - a. 100
 - b. 110
 - c. 21
 - d. 90

NARRBEGIN: Exhibit 12-1

Exhibit 12-1

When individuals in a sample of 150 were asked whether or not they supported capital punishment, the following information was obtained.

Do you support capital punishment?	Number of individuals
Yes	40
No	60
No Opinion	50

We are interested in determining whether or not the opinions of the individuals (as to Yes, No, and No Opinion) are uniformly distributed.

NARREND

11. Refer to Exhibit 12-1. The expected frequency for each group is
 - a. 0.333
 - b. 0.50
 - c. $1/3$
 - d. 50
12. Refer to Exhibit 12-1. The calculated value for the test statistic equals
 - a. 2
 - b. -2
 - c. 20
 - d. 4
13. Refer to Exhibit 12-1. The number of degrees of freedom associated with this problem is
 - a. 150
 - b. 149
 - c. 2
 - d. 3

14. Refer to Exhibit 12-1. The p -value is
- larger than 0.1
 - less than 0.1
 - less than 0.05
 - larger than 0.9
15. Refer to Exhibit 12-1. The conclusion of the test (at 95% confidence) is that the
- distribution is uniform
 - distribution is not uniform
 - test is inconclusive
 - None of these alternatives is correct.

NARRBEGIN: Exhibit 12-2

Exhibit 12-2

Last school year, the student body of a local university consisted of 30% freshmen, 24% sophomores, 26% juniors, and 20% seniors. A sample of 300 students taken from this year's student body showed the following number of students in each classification.

Freshmen	83
Sophomores	68
Juniors	85
Seniors	64

We are interested in determining whether or not there has been a significant change in the classifications between the last school year and this school year.

NARREND

16. Refer to Exhibit 12-2. The expected number of freshmen is
- 83
 - 90
 - 30
 - 10
17. Refer to Exhibit 12-2. The expected frequency of seniors is
- 60
 - 20%
 - 68
 - 64
18. Refer to Exhibit 12-2. The calculated value for the test statistic equals
- 0.5444
 - 300
 - 1.6615
 - 6.6615
19. Refer to Exhibit 12-2. The p -value is
- less than .005
 - between .025 and 0.05
 - between .05 and 0.1
 - greater than 0.1

20. Refer to Exhibit 12-2. At 95% confidence, the null hypothesis
- should not be rejected
 - should be rejected
 - was designed wrong
 - None of these alternatives is correct.

NARRBEGIN: Exhibit 12-3

Exhibit 12-3

In order to determine whether or not a particular medication was effective in curing the common cold, one group of patients was given the medication, while another group received sugar pills. The results of the study are shown below.

	Patients Cured	Patients Not Cured
Received medication	70	10
Received sugar pills	20	50

We are interested in determining whether or not the medication was effective in curing the common cold.

NARREND

21. Refer to Exhibit 12-3. The expected frequency of those who received medication and were cured is
- 70
 - 150
 - 28
 - 48
22. Refer to Exhibit 12-3. The test statistic is
- 10.08
 - 54.02
 - 1.96
 - 1.645
23. Refer to Exhibit 12-3. The number of degrees of freedom associated with this problem is
- 4
 - 149
 - 1
 - 3
24. Refer to Exhibit 12-3. The hypothesis is to be tested at the 5% level of significance. The critical value from the table equals
- 3.84
 - 7.81
 - 5.99
 - 9.34
25. Refer to Exhibit 12-3. The p -value is
- less than .005
 - between .005 and .01
 - between .01 and .025
 - between .025 and .05

NARRBEGIN: Exhibit 12-4

Exhibit 12-4

In the past, 35% of the students at ABC University were in the Business College, 35% of the students were in the Liberal Arts College, and 30% of the students were in the Education College. To see whether or not the proportions have changed, a sample of 300 students was taken. Ninety of the sample students are in the Business College, 120 are in the Liberal Arts College, and 90 are in the Education College.

NARREND

26. Refer to Exhibit 12-4. This problem is an example of a
 - a. normally distributed variable
 - b. test for independence
 - c. Poisson distributed variable
 - d. multinomial population
27. Refer to Exhibit 12-4. The expected frequency for the Business College is
 - a. 0.3
 - b. 0.35
 - c. 90
 - d. 105
28. Refer to Exhibit 12-4. The calculated value for the test statistic equals
 - a. 0.01
 - b. 0.75
 - c. 4.29
 - d. 4.38
29. Refer to Exhibit 12-4. The hypothesis is to be tested at the 5% level of significance. The critical value from the table equals
 - a. 1.645
 - b. 1.96
 - c. 5.991
 - d. 7.815
30. Refer to Exhibit 12-4. The p -value is
 - a. greater than 0.1
 - b. between 0.05 and 0.1
 - c. between 0.025 and 0.05
 - d. between 0.01 and .025
31. Refer to Exhibit 12-4. The conclusion of the test is that the
 - a. proportions have changed significantly
 - b. proportions have not changed significantly
 - c. test is inconclusive
 - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 12-5

Exhibit 12-5

The table below gives beverage preferences for random samples of teens and adults.

	Teens	Adults	Total
Coffee	50	200	250
Tea	100	150	250
Soft Drink	200	200	400
Other	<u>50</u>	<u>50</u>	<u>100</u>
400	400	600	1,000

We are asked to test for independence between age (i.e., adult and teen) and drink preferences.
NARREND

32. Refer to Exhibit 12-5. With a .05 level of significance, the critical value for the test is
 - a. 1.645
 - b. 7.815
 - c. 14.067
 - d. 15.507
33. Refer to Exhibit 12-5. The expected number of adults who prefer coffee is
 - a. 0.25
 - b. 0.33
 - c. 150
 - d. 200
34. Refer to Exhibit 12-5. The test statistic for this test of independence is
 - a. 0
 - b. 8.4
 - c. 62.5
 - d. 82.5
35. Refer to Exhibit 12-5. The p -value is
 - a. between .1 and .05
 - b. between .05 and .025
 - c. between .025 and .01
 - d. less than 0.005

NARRBEGIN: Exhibit 12-6
Exhibit 12-6

The owner of a car wash wants to see if the arrival rate of cars follows a Poisson distribution. In order to test the assumption of a Poisson distribution, a random sample of 150 ten-minute intervals was taken. You are given the following observed frequencies:

Number of Cars Arriving in a 10-Minute Interval	Frequency
0	3
1	10
2	15
3	23
4	30
5	24
6	20
7	13
8	8
9	<u>4</u>
	150

NARREND

36. Refer to Exhibit 12-6. The expected frequency of exactly 3 cars arriving in a 10-minute interval is
 - a. 0.1533
 - b. 0.1743
 - c. 23
 - d. 26.145
37. Refer to Exhibit 12-6. The calculated value for the test statistic equals
 - a. -0.18
 - b. 0.18
 - c. 1.72
 - d. 2.89
38. Refer to Exhibit 12-6. The p -value is
 - a. greater than 0.1
 - b. between 0.05 and 0.1
 - c. between .025 and .05
 - d. between .01 and .025
39. Refer to Exhibit 12-6. At 95% confidence, the conclusion of the test is that the
 - a. arrival of cars follows a Poisson distribution
 - b. arrival of cars does not follow a Poisson distribution
 - c. test is inconclusive
 - d. None of these alternatives is correct.

NARRBEGIN: Exhibit 12-7

Exhibit 12-7

You want to test whether or not the following sample of 30 observations follows a normal distribution. The mean of the sample equals 11.83, and the standard deviation equals 4.53.

2	3	5	5	7	8	8	9	9	10
11	11	12	12	12	12	13	13	13	14
15	15	15	16	16	17	17	18	18	19

NARREND

40. Refer to Exhibit 12-7. The number of intervals or categories used to test the hypothesis for this problem is
 - a. 4
 - b. 5
 - c. 6
 - d. 10
41. Refer to Exhibit 12-7. The expected frequency in the 3rd interval is
 - a. 3
 - b. 4
 - c. 5
 - d. 10

42. Refer to Exhibit 12-7. The calculated value for the test statistic equals
- 0
 - 1.67
 - 2
 - 6
43. Refer to Exhibit 12-7. The p -value is
- greater than 0.1
 - between .05 and 0.1
 - between .025 and .05
 - between .01 and .025
44. Refer to Exhibit 12-7. At 95% confidence, the conclusion of the test is that the
- data follows a normal distribution
 - data does not follow a normal distribution
 - test is inconclusive
 - None of these alternatives is correct.

NARRBEGIN: Exhibit 12-8

Exhibit 12-8

The following shows the number of individuals in a sample of 300 who indicated they support the new tax proposal.

Political Party	Support
Democrats	100
Republicans	120
Independents	80

We are interested in determining whether or not the opinions of the individuals of the three groups are uniformly distributed.

NARREND

45. Refer to Exhibit 12-8. The expected frequency for each group is
- 0.333
 - 0.50
 - 50
 - None of these alternatives is correct.
46. Refer to Exhibit 12-8. The calculated value for the *test statistic* equals
- 300
 - 4
 - 0
 - 8
47. Refer to Exhibit 12-8. The number of degrees of freedom associated with this problem is
- 2
 - 3
 - 300
 - 299

PROBLEM

1. In the last presidential election, before the candidates started their major campaigns, the percentages of registered voters who favored the various candidates were as follows.

	Percentages
Republicans	34%
Democrats	43%
Independents	23%

After the major campaigns began, a random sample of 400 voters showed that 172 favored the Republican candidate; 164 were in favor of the Democratic candidate; and 64 favored the Independent candidate. We are interested in determining whether the proportion of voters who favored the various candidates had changed.

- a. Compute the test statistic.
 - b. Using the p -value approach, test to see if the proportions have changed.
 - c. Using the critical value approach, test the hypotheses.
2. During the first few weeks of the new television season, the evening news audience proportions were recorded as ABC- 31%, CBS- 34%, and NBC- 35%. A sample of 600 homes yielded the following viewing audience data.

	Number of Homes
ABC	150
CBS	200
NBC	250

We want to determine whether or not there has been a significant change in the number of viewing audience of the three networks.

- a. State the null and alternative hypotheses to be tested.
 - b. Compute the expected frequencies.
 - c. Compute the test statistic.
 - d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
 - e. Determine the p -value and perform the test.
3. The results of a recent study regarding smoking and three types of illness are shown in the following table.

Illness	Non-Smokers	Smokers	Totals
Emphysema	20	60	80
Heart problem	70	80	150
Cancer	30	40	70
Totals	120	180	300

We are interested in determining whether or not illness is independent of smoking.

- a. State the null and alternative hypotheses to be tested.
 - b. Show the contingency table of the expected frequencies.
 - c. Compute the test statistic.
 - d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test. What do you conclude?
 - e. Determine the p -value and perform the test.
4. Among 1,000 managers with degrees in business administration, the following data have been accumulated as to their fields of concentration.

Major	Top Management	Middle Management	TOTAL
Management	280	220	500
Marketing	120	80	200
Accounting	150	150	300
TOTAL	550	450	1000

We want to determine if the position in management is independent of field (major) of concentration.

- a. Compute the test statistic.
 - b. Using the p -value approach at 90% confidence, test to determine if management position is independent of major.
 - c. Using the critical value approach, test the hypotheses. Let $\alpha = 0.10$.
5. From a poll of 800 television viewers, the following data have been accumulated as to their levels of education and their preference of television stations. We are interested in determining if the selection of a TV station is independent of the level of education.

	Educational Level			TOTAL
	High School	Bachelor	Graduate	
Public Broadcasting	50	150	80	280
Commercial Stations	150	250	120	520
TOTAL	200	400	200	800

- a. State the null and the alternative hypotheses.
 - b. Show the contingency table of the expected frequencies.
 - c. Compute the test statistic.
 - d. The null hypothesis is to be tested at 95% confidence. Determine the critical value for this test.
 - e. Determine the p -value and perform the test.
6. Before the start of the Winter Olympics, it was expected that the percentages of medals awarded to the top contenders to be as follows.

	Percentages
United States	25%
Germany	22%
Norway	18%
Austria	14%
Russia	11%
France	10%

Midway through the Olympics, of the 120 medals awarded, the following distribution was observed.

	Number of Medals
United States	33
Germany	36
Norway	18
Austria	15
Russia	12
France	6

We want to test to see if there is a significant difference between the expected and actual awards given.

- Compute the test statistic.
 - Using the p -value approach, test to see if there is a significant difference between the expected and the actual values. Let $\alpha = .05$.
 - At 95% confidence, test for a significant difference using the critical value approach.
7. A medical journal reported the following frequencies of deaths due to cardiac arrest for each day of the week:

	Cardiac Death by Day of the Week
Day	f
Monday	40
Tuesday	17
Wednesday	16
Thursday	29
Friday	15
Saturday	20
Sunday	17

We want to determine whether the number of deaths is uniform over the week.

- Compute the test statistic.
 - Using the p -value approach at 95% confidence, test for the uniformity of death over the week.
 - Using the critical value approach, perform the test for uniformity.
8. Before the presidential debates, it was expected that the percentages of registered voters in favor of various candidates would be as follows.

	Percentages
Democrats	48%
Republicans	38%
Independent	4%
Undecided	10%

After the presidential debates, a random sample of 1200 voters showed that 540 favored the Democratic candidate; 480 were in favor of the Republican candidate; 40 were in favor of the Independent candidate, and 140 were undecided. We want to see if the proportion of voters has changed.

- Compute the test statistic.
- Use the p -value approach to test the hypotheses. Let $\alpha = .05$.
- Using the critical value approach, test the hypotheses. Let $\alpha = .05$.

- Last school year, in the school of Business Administration, 30% were Accounting majors, 24% Management majors, 26% Marketing majors, and 20% Economics majors. A sample of 300 students taken from this year's students of the school showed the following number of students in each major:

Accounting	83
Management	68
Marketing	85
Economics	<u>64</u>
Total	300

We want to see if there has been a significant change in the number of students in each major.

- Compute the test statistic.
- Has there been any significant change in the number of students in each major between the last school year and this school year. Use the p -value approach and let $\alpha = .05$.

- The personnel department of a large corporation reported sixty resignations during the last year. The following table groups these resignations according to the season in which they occurred:

Season	Number of Resignations
Winter	10
Spring	22
Summer	19
Fall	9

Test to see if the number of resignations is uniform over the four seasons.

Let $\alpha = 0.05$.

- In 2002, forty percent of the students at a major university were Business majors, 35% were Engineering majors and the rest of the students were majoring in other fields. In a sample of 600 students from the same university taken in 2003, two hundred were Business majors, 220 were Engineering majors and the remaining students in the sample were majoring in other fields. At 95% confidence, test to see if there has been a significant change in the proportions between 2002 and 2003.
- Before the rush began for Christmas shopping, a department store had noted that the percentage of its customers who use the store's credit card, the percentage of those who use a major credit card, and the percentage of those who pay cash are the same. During the Christmas rush in a sample of 150 shoppers, 46 used the store's credit card; 43 used a major credit card; and 61 paid cash. With $\alpha = 0.05$, test to see if the methods of payment have changed during the Christmas rush.

13. A major automobile manufacturer claimed that the frequencies of repairs on all five models of its cars are the same. A sample of 200 repair services showed the following frequencies on the various makes of cars.

Model of Car	Frequency
A	32
B	45
C	43
D	34
E	46

At $\alpha = 0.05$, test the manufacturer's claim.

14. A lottery is conducted that involves the random selection of numbers from 0 to 4. To make sure that the lottery is fair, a sample of 250 was taken. The following results were obtained:

Value	Frequency
0	40
1	45
2	55
3	60
4	50

- State the null and alternative hypotheses to be tested.
 - Compute the test statistic.
 - The null hypothesis is to be tested at the 5% level of significance. Determine the critical value from the table.
 - What do you conclude about the fairness of this lottery?
15. The makers of Compute-All know that in the past, 40% of their sales were from people under 30 years old, 45% of their sales were from people who are between 30 and 50 years old, and 15% of their sales were from people who are over 50 years old. A sample of 300 customers was taken to see if the market shares had changed. In the sample, 100 of the people were under 30 years old, 150 people were between 30 and 50 years old, and 50 people were over 50 years old.
- State the null and alternative hypotheses to be tested.
 - Compute the test statistic.
 - The null hypothesis is to be tested at the 1% level of significance. Determine the critical value from the table.
 - What do you conclude?
16. The following table shows the results of recent study regarding gender of individuals and their selected field of study.

Field of study	Male	Female	TOTAL
Medicine	80	40	120
Business	60	20	80
Engineering	160	40	200
TOTAL	300	100	400

We want to determine if the selected field of study is independent of gender.

- Compute the test statistic.
- Using the p -value approach at 90% confidence, test to see if the field of study is independent of gender.
- Using the critical method approach at 90% confidence, test for the independence of major and gender.

17. Shown below is 3 x 2 contingency table with observed values from a sample of 1,500. At 95% confidence, test for independence of the row and column factors.

Row Factor	Column Factor		Total
	X	Y	
A	450	300	750
B	300	300	600
C	100	50	150
Total	850	650	1,500

18. Shown below is 2 x 3 contingency table with observed values from a sample of 500. At 95% confidence using the critical value approach, test for independence of the row and column factors.

Row Factor	Column Factor		
	X	Y	Z
A	40	50	110
B	60	100	140

19. A sample of 150 individuals (males and females) was surveyed, and the individuals were asked to indicate their yearly incomes. Their incomes were categorized as follows.

Category 1	\$20,000	up to	\$40,000
Category 2	\$40,000	up to	\$60,000
Category 3	\$60,000	up to	\$80,000

The results of the survey are shown below.

Income Category	Male	Female
Category 1	10	30
Category 2	35	15
Category 3	15	45

We want to determine if yearly income is independent of gender.

- Compute the test statistic.
- Using the p -value approach, test to determine if yearly income is independent of gender.

20. A group of 2000 individuals from 3 different cities were asked whether they owned a foreign or a domestic car. The following contingency table shows the results of the survey.

Type of Car	CITY			Total
	Detroit	Atlanta	Denver	
Domestic	80	200	520	800
Foreign	<u>120</u>	<u>600</u>	<u>480</u>	<u>1200</u>
Total	200	800	1000	2000

At $\alpha = 0.05$ using the p -value approach, test to determine if the type of car purchased is independent of the city in which the purchasers live.

21. Dr. Sherri Brock's diet pills are supposed to cause significant weight loss. The following table shows the results of a recent study where some individuals took the diet pills and some did not.

	Diet Pills	No Diet Pills	Total
No Weight Loss	80	20	100
Weight Loss	100	100	200
Total	180	120	300

We want to see if losing weight is independent of taking the diet pills.

- Compute the test statistic.
 - Using the p -value approach at 95% confidence, test to determine if weight loss is independent on taking the pill.
 - Use the critical method approach and test for independence.
22. Five hundred randomly selected automobile owners were questioned on the main reason they had purchased their current automobile. The results are given below.

	Styling	Engineering	Fuel Economy	Total
Male	70	130	150	350
Female	<u>30</u>	<u>20</u>	<u>100</u>	<u>150</u>
Total	100	150	250	500

- State the null and alternative hypotheses for a contingency table test.
 - State the decision rule for the critical value approach. Let $\alpha = .01$.
 - Calculate the χ^2 test statistic.
 - Give your conclusion for this test.
23. A group of 500 individuals were asked to cast their votes regarding a particular issue of the Equal Rights Amendment. The following contingency table shows the results of the votes:

Sex	Favor	Undecided	Oppose	TOTAL
Female	180	80	40	300
Male	150	20	30	200
TOTAL	330	100	70	500

At $\alpha = .05$ using the p -value approach, test to determine if the votes cast were independent of the sex of the individuals.

24. Two hundred fifty managers with degrees in business administration indicated their fields of concentration as shown below.

Major	Top Management	Middle Management	TOTAL
Management	65	60	125
Marketing	30	20	50
Accounting	25	50	75
TOTAL	120	130	250

At $\alpha = .01$ using the p -value approach, test to determine if the position in management is independent of the major of concentration.

25. From a poll of 800 television viewers, the following data have been accumulated as to their levels of education and their preference of television stations.

	Level of Education			
	High School	Bachelor	Graduate	TOTAL
Public Broadcasting	110	190	100	400
Commercial Stations	80	220	100	400
TOTAL	190	410	200	800

Test at $\alpha = .05$ to determine if the selection of a TV station is dependent upon the level of education. Use the p -value approach.

26. The data below represents the fields of specialization for a randomly selected sample of undergraduate students. We want to determine whether there is a significant difference in the fields of specialization between regions of the country.

	Northeast	Midwest	South	West	Total
Business	54	65	28	93	240
Engineering	15	24	8	33	80
Liberal Arts	65	84	33	98	280
Fine Arts	13	15	7	25	60
Health Sciences	<u>3</u>	<u>12</u>	<u>4</u>	<u>21</u>	<u>40</u>
	150	200	80	270	700

- Determine the critical value of the chi-square χ^2 for this test of independence.
 - Calculate the value of the test statistic.
 - What is the conclusion for this test? Let $\alpha = .05$.
27. A department store believes that telephone calls come into the switchboard at 10-minute intervals, according to a Poisson distribution. Before ordering new equipment, the store wishes to determine whether the Poisson model is a valid assumption. Records on the number of calls received were kept for a random selection of 150 ten-minute intervals. The results are shown below.

Number of Calls	Frequency
0	5
1	18
2	24
3	30
4	32
5	13
6	20
7	<u>8</u>
	150

- a. What is the average number of calls during these ten-minute intervals?
- b. Generate the expected number of calls using a Poisson probability table.
- c. Give the null and alternative hypotheses for the appropriate test.
- d. Determine the number of degrees of freedom for this test.
- e. Calculate the value of the test statistic.
- f. Determine the p -value and state whether or not the Poisson model is a valid model for the phone calls?

28. It is believed that sales of books at a local bookstore follow a Poisson distribution. Below you are given information on the number of books sold during a sample of 15-minute intervals.

Number of Books	Frequency
0	2
1	3
2	12
3	16
4	19
5	20
6	18
7	16
8	9
9	<u>5</u>
	120

- a. What is the average number of books sold during a 15-minute period?
- b. Using the Poisson distribution, find the probability associated with the number of books sold.
- c. Generate the expected number of books sold using a Poisson probability table.
- d. State the null and alternative hypotheses.
- e. Calculate the test statistic.
- f. Use the p -value approach to test the hypotheses. What is your conclusion? Let $\alpha = .05$.

29. The number of emergency calls per day at a hospital over a period of 120 days is shown below.

Number of Emergency Calls (x)	Observed Frequency (f)
0	9
1	12
2	30
3	27
4	22
5	13
6	<u>7</u>
Total	120

Use $\alpha = 0.05$ and the p -value approach to see if the above data have a Poisson distribution.

30. An insurance company has gathered the following information regarding the number of accidents reported per day over a period of 100 days.

Accidents Per Day	Number of Days (f)
0	5
1	18
2	25
3	24
4	20
5	8

Using the critical value approach test to see if the above data have a Poisson distribution. Let $\alpha = 0.05$.

31. A professor believes that the final examination scores in statistics are normally distributed. A sample of 40 final scores has been taken. You are given the sample below. The mean of the scores is 83.1, and the standard deviation is 10.43.

56	63	65	68	72	72	73	75	77	78
78	79	80	80	80	80	80	80	81	81
82	84	84	86	86	87	88	90	90	92
92	93	93	94	95	96	97	98	100	100

- State the null and alternative hypotheses.
 - Compute the test statistic for the goodness of fit test.
 - At 99% confidence using the p -value approach, test the hypotheses. What do you conclude about the distribution of final examination scores?
32. A manager believes that the shelf life of apple juice is normally distributed. A sample of 30 containers of juice was taken and the shelf life was recorded. You are given the results below. The average shelf life in the sample was 23.07 days with a standard deviation of 4.29 days.

15	17	19	20	20	20	21	21	21	21
21	21	21	22	22	22	22	22	22	22
24	24	25	25	27	29	30	31	32	33

- State the null and alternative hypotheses.
 - Compute the test statistic for the goodness of fit test.
 - At 95% confidence using the p -value approach, test the hypotheses. What do you conclude about the distribution?
33. The following data show the grades of a sample of 40 students who have taken statistics.

98	64	96	69
45	94	58	59
63	49	88	83
85	87	68	77
56	63	86	89
84	73	52	63
64	80	69	68
79	73	78	79
72	82	78	88
83	76	66	76

Use $\alpha = 0.1$ and conduct a goodness of fit test to determine if the sample comes from a population that has a normal distribution. Use the critical value approach.

34. Use $\alpha = 0.05$ to determine if the following sample comes from a normal distribution. Use the critical value approach.

105	260	314	400	520
300	306	115	200	208
418	110	410	312	360
310	314	418	316	412
516	480	490	504	518
280	270	516	419	520
420	438	511	708	300
420	519	702	690	518
510	700	650	670	612
460	600	680	692	600

35. We want to determine if the following sample comes from a normal distribution.

105	260	314	400	520
300	306	115	200	208
418	110	410	312	360
310	314	418	316	412
516	480	490	504	518
280	270	516	419	520
420	438	511	708	300
420	519	702	690	518
510	700	650	670	612
460	600	680	692	600

- Compute the mean and the standard deviation.
- Compute the test statistic. Hint: divide the distribution into 10 equal intervals.
- At 95% confidence using the critical value approach, test to determine if the sample comes from a normal population.
- Compute the p -value.